MIL-C-48203(PA)
30 October 1972

MILITARY SPECIFICATION

CRANK ASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of crank assembly used on T4090 test set.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 ----- Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105 --- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 -- Lot Numbering of Ammunition
MIL-STD-1169 -- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1190

DRAWINGS

ORDNANCE CORPS

8808362 ----- Packing and Marking for Storage and Shipments
8834345 ----- Crank Assembly
8834386 ----- Yoke, Crank
8883537 ----- Bag, Packing for Repair Parts
EL-8834386 -- Equipment List

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The crank assembly shall comply with all requirements specified on the drawing (dwg) 8834345, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Workmanship.

3.3.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.3.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.3.3 Burrs. Parts shall be free from burrs.

3.3.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.3.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requiremens as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I and Table II), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table I

Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table II

Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Yoke, crank (see dwg 8834386) covering a detail of dwg 8834345.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major		
101.	Pitch diameter of threaded side hole	Gage
102.	Minor diameter of threaded side hole	Gage
103.	Distance from flat to circumference in shaft hole (.108)	Gage
104.	Material (see 3.1)	Visual
Minor		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.2 Crank assembly (see dwg 8834345).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major		
101.	Detent works properly	Manual
102.	Grip turns freely	Manual
103.	Crank operates freely with respect to yoke thru 180 degrees	Manual
Minor		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.3 Preparation for delivery (see 5.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major		
101.	Preservation, packaging, packing and marking in accordance with MIL-STD-1169	Visual/Manual
Minor:	None defined.	

4.3.2.1.4 Bag, packing for repair parts (see dwg 8883537) covering Part Number 8883537-6 and 5.2.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major		
101.	Material (see 3.1)	Visual
102.	Length before packing, min.	*Measure
103.	Width before packing, min.	*Measure
104.	Properly packed	Visual/Manual
105.	Proper heat seal	Visual/Manual
106.	Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116	Visual
107.	No tear or hole in bag	Visual
Minor		
201.	Marking correct and legible	Visual
202.	Workmanship (see 3.3)	Visual/Manual

*Measure with suitable scale.

4.3.2.1.5 Packing and marking for storage and shipments (see dwg 8808362) and 5.2.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major		
101.	Gross load weight, max.	*Measure
102.	Contents tight in package	Manual
103.	Correctly packed	Visual/Manual
104.	Material (see 3.1)	Visual
Minor		
201.	Marking correct and legible	Visual
202.	Workmanship (see 3.3)	Visual/Manual

*Determine weight with suitable scale.

4.3.3 Testing. Not applicable.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL's) (see EL-8834386). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring activity.

4.4 Test methods. Not applicable.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The crank assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The crank assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The crank assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1).

5.2.1.1 Level A. The crank assembly shall be packaged in accordance with dwg 8883537 and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. The crank assembly shall be packaged in accordance with dwg 8808362, except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Levels A and C. The crank assembly shall be packed in accordance with dwg 8883537 and 8808362.

5.2.3 Marking. Marking shall be as specified in accordance with dwg 8883537 and 8808362, as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Production or repair parts.

c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-N-I.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A063

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802